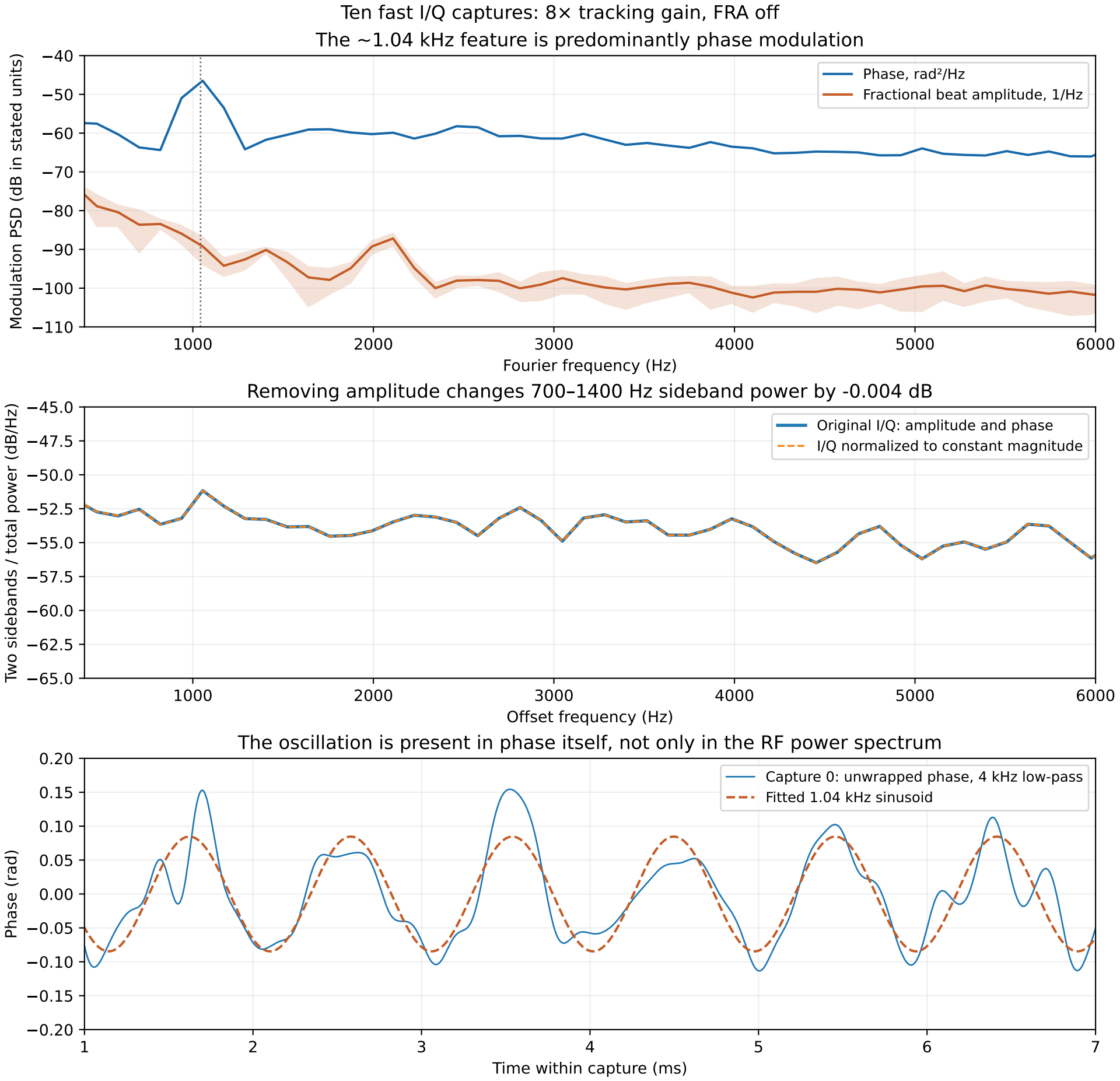

Ten fast I/Q captures: 8× tracking gain, FRA off
The ~1.04 kHz feature is predominantly phase modulation
Phase, rad²/Hz
Fractional beat amplitude, 1/Hz
Modulation PSD (dB in stated units)
Fourier frequency (Hz)
Removing amplitude changes 700–1400 Hz sideband power by -0.004 dB
Original I/Q: amplitude and phase
I/Q normalized to constant magnitude
Two sidebands / total power (dB/Hz)
Offset frequency (Hz)
The oscillation is present in phase itself, not only in the RF power spectrum
Capture 0: unwrapped phase, 4 kHz low-pass
Fitted 1.04 kHz sinusoid
Phase (rad)
Time within capture (ms)
Welch bin spacing 117.19 Hz; 17.07 ms records. Shading: 10–90% record spread.
No attribution to individual lasers or electronics; captures need not coincide with analyzer traces.